Belief Propagation Decoding of Primitive Rateless Codes

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***Abstract*— In this letter, we propose a novel decoding scheme for primitive rateless (PR) codes to improve belief propagation (BP) performance in the short blocklength regime. We first design the PR codes using primitive polynomials whose supports form a Golomb ruler, enabling the construction of a 4-cycle-free parity-check matrix (PCM) without the need for auxiliary variable nodes. We introduce a second PCM derived by sorting channel log-likelihood ratios and applying Gaussian elimination. The double-bases decoder operates by running BP decoding in parallel on both PCMs, and selecting the candidate codeword with the shorter Euclidean distance to the received signal. Simulation results show that the proposed double-bases BP approach can outperform the multiple-bases BP decoder while reducing the number of required PCMs and complexity. The PR code paired with the proposed decoder offers competitive performance compared to 5G-polar, RM, and LDPC codes in the short blocklength regime.**

***Index Terms*— Belief propagation, Golomb ruler, primitive rateless codes, ultra-reliable low-latency communications.**

I. Introduction

EMERGING beyond-5G (B5G) and 6G networks pose stringent requirements for wireless systems, particularly in ultra-reliable low-latency communications (URLLC) and massive machine-type communications (mMTC). URLLC demands sub-millisecond latency with frame error rates as low as 10^{-5} to 10^{-7} [1], while mMTC targets reliable, energy-efficient transmission of short packets from a large number of low-power devices. These use cases require short codes that offer high reliability and low decoding latency and complexity [2].

Algebraic codes, such as Reed–Muller (RM) [3] and Bose–Chaudhuri–Hocquenghem (BCH) [4] codes, have recently regained attention due to their strong minimum distance properties [5]. To approach maximum-likelihood (ML) decoding, universal algorithms such as ordered statistics decoding (OSD) [6] and guessing random additive noise decoding (GRAND) [7] have been proposed. However, these methods become computationally intensive at low SNRs and rates typical of URLLC and mMTC scenarios [8], [9]. Belief propagation (BP) offers a lower-complexity, parallelizable alternative [10], but its effectiveness depends on the sparsity and cycle structure of the parity-check matrix (PCM) [11]. Many algebraic codes have dense PCMs with short cycles, notably 4-cycles, that degrade BP performance. Several improvements have been explored: adaptive BP (ABP) [12], which permutes the PCM based on channel reliabilities using Gaussian elimination (GE) per iteration; and structural design techniques such as low-weight parity-check (LWPC) matrices [13] or 4-cycle elimination using auxiliary variables [14]. However, auxiliary-node-based designs convey no information and can degrade performance, particularly in AWGN channels. More recently, the progressive row growth (PRG) scheme [15] enables 4-cycle-free PCM construction with improved connectivity and minimal auxiliary overhead. Yet, near-ML performance still often requires multiple-bases BP (MBBP) decoding [16], which relies on constructing several diverse PCMs from low-weight dual codewords, an approach that is computationally expensive and hard to scale. Primitive rateless (PR) codes [17] offer a promising solution. Generated from m-sequences based on primitive polynomials over $\mathbf{GF}(2)$, PR codes support natural rate compatibility and high flexibility. When constructed using Golomb ruler-supported primitive polynomials, PR codes can inherently produce 4-cycle-free PCMs without auxiliary variables [18], enabling efficient BP decoding.

In this letter, we propose a low-complexity double-bases BP decoder for PR codes. The scheme employs two PCMs: a sparse 4-cycle-free PCM derived from a Golomb ruler, and a second matrix generated by sorting channel log-likelihood ratios (LLRs) followed by GE. Unlike ABP, the second PCM is fixed and computed only once, significantly reducing complexity. The double-bases decoder operates by running BP decoding in parallel on both PCMs, producing two candidate codewords. The correct codeword is then selected as the codeword with the minimum Euclidean distance to the received signal. Simulation results show that this approach can outperform MBBP while offering a practical solution for decoding short PR codes.

II. Preliminaries

A. Primitive Rateless (PR) Codes

A binary PR code, denoted as $\mathrm{PR}(k, p(x))$, is characterized by the information block length k and a binary primitive polynomial $p(x) = \sum_{i=0}^{k} p_i x^i$ of degree k, where $p_0 = p_k = 1$. This polynomial is the minimal polynomial of a primitive element α over $\mathbf{GF}(2^k)$. A PR code truncated to length n is denoted by $\mathrm{PR}(k, n, p(x))$ and its generator matrix is constructed as follows [17]

$$\mathbf{G} = \left[\alpha^0, \alpha^1, \ldots, \alpha^{n-1}\right], \tag{1}$$

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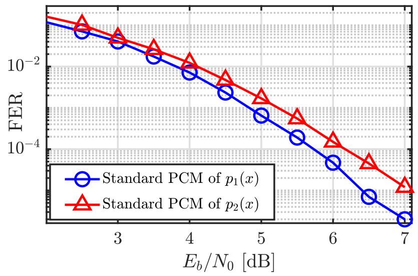


Fig. 1. FER performance of rate-$1/2$PR$(16, 32, p(x))$ code constructed using $p_1(x)$ and $p_2(x)$, under BP decoding with $I_{max} = 40$.

where the i^{th} column ($1 \leq i \leq n$) is the binary representation of α^{i-1}. The standard PCM of PR$(k, n, p(x))$ is given by [17]

$$\mathbf{H} = \begin{bmatrix} p_0 & p_1 & \dots & p_k & 0 & 0 & \dots \\ 0 & p_0 & p_1 & \dots & p_k & 0 & \dots \\ & \ddots & & \ddots & & \ddots & \\ 0 & \dots & 0 & p_0 & p_1 & \dots & p_k \end{bmatrix}, \quad (2)$$

where the i^{th} row is the $(i-1)^{th}$-order cyclic shift of the first row, for $2 \leq\ i \leq\ n-k$.

B. Requirements for the 4-Cycle Free PCM

For PR codes, it has been shown that, with an appropriate choice of the primitive polynomial, it is possible to construct a code with a PCM free of length-4 cycles [18]. In particular, as demonstrated in [18, Theorem 1], a necessary and sufficient condition for the PCM of a PR code, based on a binary primitive polynomial of the form $p(x) = 1+p_1x+\ldots+p_{k-1}x^{k-1}+p_kx^k$, as shown in (2), to be free of length-4 cycles is that the support set of $p(x)$ forms a Golomb ruler. The support of $p(x)$ is defined as $\mathcal{S}_p = \{1 \leq i \leq k | p_i = 1\}$ and a set of integers $\mathcal{A} = \{a_1, a_2, \ldots, a_m\}$ where $a_1 < a_2 < \ldots < a_m$ is called a Golomb ruler if for each integer $x \neq 0$ there is at most one solution to equation $x = a_j - a_i$ for $a_j, a_i \in \mathcal{A}$. This property holds independently of the code rate; that is, regardless of the selected codeword length, a PR code constructed with a Golomb ruler-supported primitive polynomial will always yield a PCM free of length-4 cycles. While our focus is on using primitive polynomials to generate PR codes, we note that irreducible but non-primitive polynomials could also be considered. Since irreducible polynomials are more abundant, they may offer a broader design space and reduced search complexity. However, unlike primitive polynomials, they do not guarantee maximal-length LFSR sequences, which may lead to earlier column repetition in the generator matrix and potentially degraded distance properties. A detailed comparison between primitive and irreducible-based PR codes remains an interesting direction for future research.

Example 1: Consider the primitive polynomial $p_1(x) = x^{16}+x^{13}+x^{12}+x^7+1$. The support vector for this polynomial is $\mathcal{S}_{p_1} = \{0, 7, 12, 13, 16\}$, which is a Golomb ruler. Thus, it has no length-4 cycles in its standard PCM at any rates. In contrast, the support vector for another primitive polynomial $p_2(x) = x^{16} + x^{12} + x^7 + x^2 + 1$ with the same weight is $\mathcal{S}_{p_2} = \{0, 2, 7, 12, 16\}$, which is not a Golomb ruler and contains 11, 27 and 75 length-4 cycles at rates $1/2$, $1/3$ and $1/6$, respectively. As illustrated in Fig. 1, the presence of these length-4 cycles leads to the frame error rate (FER) performance degradation under BP decoding.

C. Belief Propagation Decoding

BP is an iterative message-passing algorithm operating on the Tanner graph of the code. It exchanges messages between variable nodes (VNs) and check nodes (CNs), typically using log-likelihood ratios (LLRs) derived from the channel. The sum-product algorithm (SPA) [19] provides the standard BP message update rules, while practical implementations often use the normalized min-sum (NMS) approximation with a scaling factor γ. To improve convergence stability, a damping factor $\lambda \in [0, 1)$ is applied. The damped NMS message from CN c to VN v at iteration ℓ is given by:

$$L_{c\to v}^{(\ell)} = \lambda L_{c\to v}^{(\ell-1)} + \gamma(1-\lambda) \min_{v' \in \mathcal{N}_c \setminus v} |L_{v'\to c}^{(\ell-1)}| \prod_{v' \in \mathcal{N}_c \setminus v} \operatorname{sgn}\left(L_{v'\to c}^{(\ell-1)}\right). \quad (3)$$

For more detailed descriptions, we refer the reader to [20], [21, Sec. III-B], and references therein.

III. Proposed Strategy

A. Code Design

To construct PR codes with PCMs free of length-4 cycles, it is necessary to identify primitive polynomials whose support sets form a Golomb ruler. The authors in [22] have proposed constructing such polynomials by searching within subsets of marks derived from a large Golomb ruler and then verifying the primitivity of each candidate polynomial. However, this method involves a high search complexity, requires extensive primitivity testing, and lacks a guarantee of success even after exhaustive search efforts. We propose a more efficient and structured approach to generate suitable primitive polynomials. The key idea is based on the observation that a primitive polynomial with degree k and maximum weight w_p whose support can potentially form a Golomb ruler, must satisfy $\binom{w_p}{2} < k$ [22]. Furthermore, since primitive polynomials must have an odd number of non-zero terms (w_p must be an odd number), the calculated weight is adjusted to the nearest lower odd number if necessary. Given this maximum allowable weight, the algorithm systematically constructs support sets by fixing the starting point at 0 and the end point at k and selecting intermediate terms to satisfy the Golomb ruler condition. The primitivity of the candidate polynomials built from these support sets is subsequently checked. If a primitive polynomial is found, it directly guarantees the absence of length-4 cycles in the corresponding PCM. This procedure is outlined in Algorithm 1. Note that we aim to determine the maximum allowable w_p, since a PR code constructed with a higher-density polynomial generally exhibits a better minimum Hamming distance [17]. A full investigation of the relationship between the weight of a primitive polynomial and its BP decoding performance is left as a direction for future work. It is worth mentioning that Golomb rulers with up to $65,000$ marks have been exhaustively tabulated in [23], covering nearly all practical values of code dimension k. It is highly unlikely that a suitable ruler subset cannot be found to meet the weight and length constraints of a primitive polynomial.

B. Decoding Based on Double-Bases Belief Propagation

While having a 4-cycle-free PCM improves BP decoding performance, it does not necessarily achieve performance close

Algorithm 1 Weight-Constrained Golomb Ruler Search for Primitive Polynomials

Input: Information block length k
Output: Primitive polynomial $p(x)$ with degree k forming a Golomb ruler support

1 Compute maximum weight w_p such that $\binom{w_p}{2} < k$
2 **if** w_p *is even* **then**
3 $\quad w_p \leftarrow w_p - 1$
4 **end**
5 Initialize support set $\mathcal{S}_p \leftarrow \{0, k\}$
6 **while** $|\mathcal{S}_p| < w_p$ **do**
7 $\quad$ Select an intermediate point, maintaining unique pairwise differences
8 $\quad$ Add the selected point to $\mathcal{S}_p$
9 **end**
10 Construct polynomial $p(x)$ from support set $\mathcal{S}_p$
11 **if** $p(x)$ *is primitive* **then**
12 $\quad$ **return** $p(x)$;
13 **else**
14 $\quad$ Modify intermediate points and go to step 6
15 **end**

to ML decoding. To bridge this gap, advanced techniques such as MBBP are often applied. However, MBBP requires the construction of multiple PCMs, typically derived from low-weight dual codewords, a process that is computationally expensive and not always scalable. Inspired by the concept of ABP decoding, we propose a more efficient approach to generate a second PCM for PR codes, without relying on dual codewords. Let $\mathbf{H_1}$ denote the standard 4-cycle-free PCM of the PR code. The second matrix, $\mathbf{H_2}$, is generated through the following steps:

1) **Sorting and Permutation:** Upon receiving the channel LLRs (L^{ch}) at decoder, a permutation π_1 is formed to sort the magnitude of L^{ch} in a descending order. Then, this permutation is applied to the columns of $\mathbf{H_1}$ to obtain a reordered matrix $\tilde{\mathbf{H}}_\mathbf{1} = \pi_1(\mathbf{H_1})$.
2) **GE:** GE is applied to $\tilde{\mathbf{H}}_\mathbf{1}$ to generate an updated PCM $\mathbf{H_2} = [\tilde{P}|I_{n-k}]$, where the $m = n-k$ least reliable bits are mapped to the sparse part of $\mathbf{H_2}$ such that every least reliable bit is connected with one CN.

The double-bases decoder operates by running BP decoding, with the maximum number of iterations $I_{\max}$, in parallel on both $\mathbf{H_1}$ and $\mathbf{H_2}$, potentially producing two candidate codewords. The correct codeword is then selected as the codeword with the minimum Euclidean distance to the received signal. Unlike ABP, where the PCM is updated at every iteration based on LLRs, the double-bases approach keeps $\mathbf{H_2}$ fixed through all BP iterations. This significantly reduces complexity. Although $\mathbf{H_2}$ is denser than $\mathbf{H_1}$, its structure is carefully designed to enhance decoding performance. In particular, the $n-k$ least reliable bits are mapped, via GE, to columns that correspond to the identity matrix. These columns are sparse and connected to exactly one check node, effectively turning those unreliable bits into degree-one variables, which are much easier for the BP algorithm to decode accurately. While BP generally performs better with sparse matrices, the targeted local sparsity in $\mathbf{H_2}$ compensates for its overall density. The decoder benefits from this structure by isolating and resolving the least reliable bits early in the iterative process. This mechanism allows the double-bases decoder to harness the global convergence properties of $\mathbf{H_1}$ and the local correction strength of $\mathbf{H_2}$, resulting in improved overall decoding performance. The general structure and density of the two PCMs constructed for $\mathrm{PR}(16, 32)$ are shown in Fig. 2 as an example.

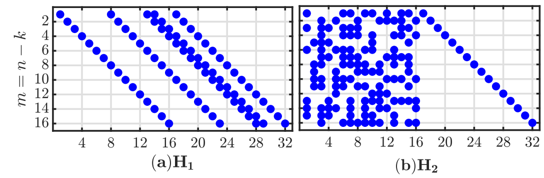


Fig. 2. Two PCMs $\mathbf{H_1}$ and $\mathbf{H_2}$ constructed for $\mathrm{PR}(16, 32)$.

C. Complexity Analysis

For simplicity, we evaluate the complexity of BP-based decoders in terms of the number of edges, $|E|$, in the PCM. The complexity of standard BP decoding is thus $\mathcal{O}(|E|I_{max})$. For MBBP with N PCMs, the complexity becomes $\mathcal{O}(\sum_{i=1}^{N}|E_i|I_{max})$. Accordingly, the computational complexity of the proposed double-bases BP decoder is given by:

$$C = \mathcal{O}(n\log n) + \mathcal{O}(n \cdot \min(k, n-k)^2) + \mathcal{O}(\sum_{i=1}^{2}|E_i|I_{\max}). \quad (4)$$

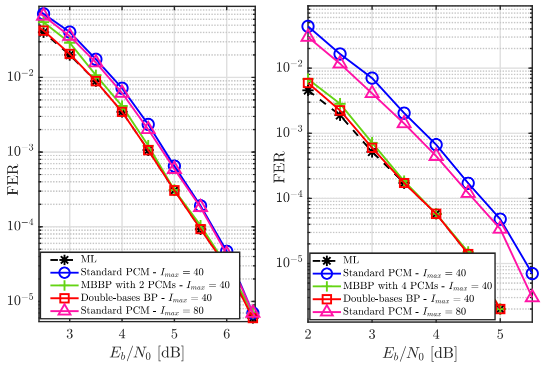
In comparison, the complexity of an order-m OSD decoder follows a similar structure in (4) but replaces the BP decoding term by $\sum_{j=1}^{m}\binom{k}{j}\mathcal{O}(k + k(n-k))$ which is the complexity of re-encoding $\sum_{j=1}^{m}\binom{k}{j}$ test error patterns.

IV. Experimental Results

In our simulations, a binary message vector $\mathbf{b}$ of length k is encoded using a PR code to produce a codeword $\mathbf{c}$. Each coded symbol c_i, for $1 \le i \le n$, is modulated using binary phase-shift keying (BPSK) as $x_i = (-1)^{c_i}$ and transmitted over a binary-input additive white Gaussian noise (BI-AWGN) channel, with noise variance $N_0/2$. For NMS decoding, all message updates are performed using equation (3) with $\gamma = 1$ and a damping factor of $\lambda = 0.7$, which were selected based on preliminary tuning. To ensure consistency in our results, the same parameters are used across all BP scheme simulations.

A. FER Performance of Very Short PR Codes

Fig. 3 presents the FER performance of two PR codes, $\mathrm{PR}(16, 32)$ and $\mathrm{PR}(24, 64)$, decoded using a BP decoder. The PR codes are constructed using the primitive polynomials $p_{16}(x) = x^{16} + x^{13} + x^{12} + x^7 + 1$ and $p_{24}(x) = x^{24} + x^{23} + x^{21} + x^{11} + 1$, respectively. As shown in Fig. 3 (a), the BP decoder applied to standard PCM $\mathrm{PR}(16, 32)$ achieves performance within 0.5 dB of ML bound and matches ML performance for FERs below 10^{-5}. We observed that increasing the number of BP iterations from 40 to 80 does not yield a significant improvement in performance. However, when decoded using the proposed double-bases BP decoder, the code meets ML performance across a wide range of SNRs. For comparison, the FER performance using MBBP with two



(a) $\mathrm{PR}(16, 32, p(x))$ (b) $\mathrm{PR}(24, 64, p(x))$

Fig. 3. (a) FER performance of a rate-1/2PR$(16, 32)$ and (b) rate-3/8PR$(24, 64)$ decoded by BP decoder.

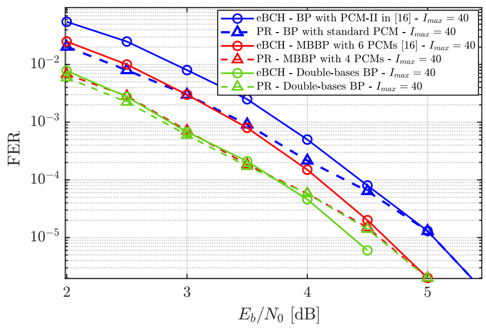


Fig. 4. FER of $\mathrm{PR}(24, 64)$ and $\mathrm{eBCH}(24, 64)$ with $I_{max} = 40$.

distinct PCMs is also shown. The second PCM for MBBP is constructed using the low-weight dual codewords, followed by optimized 4-cycle removal based on the method in [15]. Readers are referred to [15] for further details. The results demonstrate that both the double-bases BP and the MBBP decoders yield comparable performance when using the same number of PCMs. However, at lower SNRs, the proposed double-bases decoder outperforms MBBP, suggesting greater robustness in challenging channel conditions. Note that the main advantage of the double-bases BP decoder over MBBP is that it avoids the need for minimum-weight dual codeword enumeration and explicit 4-cycle removal during PCM construction. The second PCM in the double-bases BP is derived directly from the initial 4-cycle-free PCM through simple sorting and GE. This makes it particularly well-suited for rateless scenarios, as it eliminates the need to redesign PCMs for different code rates. Fig. 3 (b) illustrates the FER performance of $\mathrm{PR}(24, 64)$. In this case, the proposed double-bases BP decoder meets the near-ML performance, whereas the MBBP decoder requires four different PCMs to reach similar performance levels. The performance gain achieved by the proposed decoder is more pronounced for the $\mathrm{PR}(24, 64)$ code than for $\mathrm{PR}(16, 32)$. This is because in very short codes like $\mathrm{PR}(16, 32)$, a single well-designed PCM is often sufficient for convergence, limiting the added value of a second basis. In contrast, $\mathrm{PR}(24, 64)$ has a longer block length and higher density, leading to an increased inter-node correlation in the Tanner graph. As a result, the second matrix $\mathbf{H_2}$, designed using channel reliabilities, becomes more effective in correcting unreliable bits.

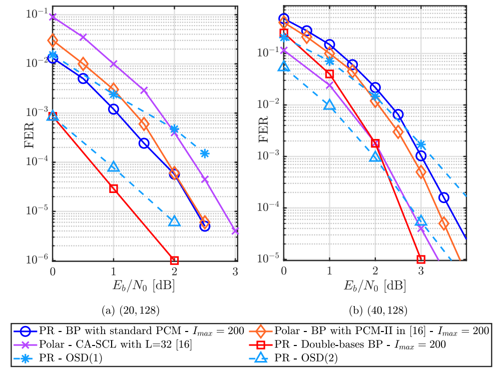


Fig. 5. (a) FER performance comparison of $\mathrm{PR}(20, 128)$ with $\mathrm{Polar}(20, 128)$ and (b) $\mathrm{PR}(40, 128)$ with $\mathrm{Polar}(40, 128)$.

B. FER Performance Comparison of PR*(24, 64) and* eBCH*(24, 64) With Redundant Rows*

The authors in [15] proposed the use of redundant parity-check rows when constructing the PCM of algebraic codes, with the aim of increasing cycle connectivity and reducing the number of small trapping sets in the Tanner graph. For comparison, we adopt the same technique to construct the PCM for $\mathrm{PR}(24, 64)$, augmenting it with redundant rows. We use low-weight dual codewords to generate the additional rows and subsequently apply a 4-cycle removal algorithm in [15] to eliminate any short cycles introduced during this process. The resulting PCM has dimensions 64×76 and contains a total of 275 edges. Fig. 4 compares the FER of $\mathrm{PR}(24, 64)$ with that of the $\mathrm{eBCH}(24, 64)$ code constructed in [15], where both codes use PCM enhanced with redundant rows. The PCM for $\mathrm{eBCH}(24, 64)$ has a significantly larger size of 272×81, with a total of 1360 edges. As shown in the figure, although both codes achieve similar performance at an FER of 10^{-5} and below, the $\mathrm{PR}(24, 64)$ code, with substantially fewer edges, exhibits superior performance at lower SNRs. This highlights the efficiency of PR codes in achieving strong error-correction performance with reduced decoding complexity. It is also shown that $\mathrm{PR}(24, 64)$ decoded by MBBP with 4 PCMs outperforms $\mathrm{eBCH}(24, 64)$ decoded with 6 PCMs. As also shown in Fig. 4, the two codes exhibit comparable performance under the proposed double-bases BP decoding, with the eBCH code outperforming the PR code at higher SNRs due to its larger minimum Hamming distance.

C. FER Performance of Short PR Codes With Length $n = 128$

Fig. 5 shows the FER of two low-rate PR codes, $\mathrm{PR}(20, 128)$ and $\mathrm{PR}(40, 128)$, under various decoding schemes, and compares the results with those of $\mathrm{Polar}(20, 128)$ and $\mathrm{Polar}(40, 128)$ codes. The standard PCMs for the PR codes are constructed using the primitive polynomials $p_{20}(x) = x^{20} + x^{18} + x^4 + x + 1$ and $p_{40}(x) = x^{40} + x^{39} + x^{35} + x^{22} + x^{10} + x^2 + 1$, respectively.

As shown in Fig. 5 (a), the $\mathrm{Polar}(20, 128)$ code, when decoded using a BP decoder with a PCM created in [15] with size 323×215 and 1224 total edges having complexity $\mathcal{O}(EI) \approx 244,800$, outperforms the CRC-aided successive cancellation list (CA-SCL) decoder with list size $L = 32$ and complexity $\mathcal{O}(Lnlogn) \approx 28,672$. However, the $\mathrm{PR}(20, 128)$

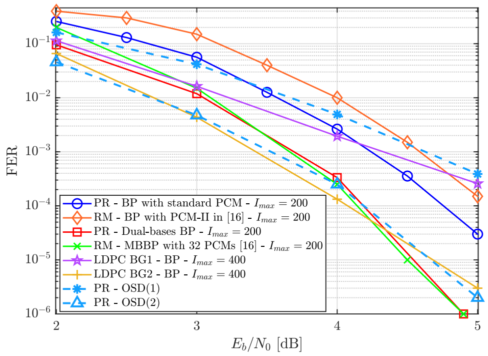


Fig. 6. FER performance of rate-$1/2 \mathrm{PR}(64, 128)$, $\mathrm{RM}(64, 128)$ and $\mathrm{LDPC}(64, 128)$.

code, using its standard PCM of size 108×128 with only 540 edges and complexity $\mathcal{O}(EI) \approx 108{,}000$, surpasses the performance of both decoding schemes. Note that although the CA-SCL is the standard decoder for polar codes, its sequential nature limits parallel implementation, resulting in higher decoding latency [24]. In contrast, BP decoding, despite a higher theoretical complexity, exhibits high inherent parallelism, making it highly suitable for hardware acceleration and low-latency applications. When the proposed double-bases BP decoder is applied to the PR code, it achieves an additional performance gain of approximately 1 dB over standard BP decoding at the cost of additional complexity incurred by GE, while also outperforming OSD(2). It is worth noting that the complexity of OSD(2) is higher than that of the double-bases BP decoder, due to the large number of re-encoding operations required in OSD. In contrast, as illustrated in Fig. 5 (b), the $\mathrm{Polar}(40, 128)$ code decoded using the CA-SCL decoder with list size $L = 32$ outperforms both the $\mathrm{Polar}(40, 128)$ and $\mathrm{PR}(40, 128)$ codes when decoded using the BP decoder. However, the $\mathrm{PR}(40, 128)$ code, when decoded using the proposed double-bases BP decoder, achieves performance that closely approaches that of the CA-SCL decoder and even surpasses it at frame error rates below 10^{-3}. It is worth noting that achieving the performance of the CA-SCL decoder for the $\mathrm{Polar}(40, 128)$ code requires the use of MBBP decoding with 32 distinct PCMs [15]. This large number of PCMs increases the total number of edges significantly and incurs higher complexity than the double-bases BP decoder. Fig. 6 compares the FER performance of $\mathrm{PR}(64, 128)$ with $\mathrm{RM}(64, 128)$ codes. As shown, the PR code with 576 edges in its PCM can outperform the RM code with 1246 edges under standard BP decoding. However, when PR code is decoded by the proposed double-bases BP decoder, an additional 1 dB gain can be achieved compared to standard BP, and its performance is comparable to LDPC BG2, which is designed for shorter block codes. In contrast, achieving comparable performance for the RM code requires the use of MBBP decoding with 32 distinct PCMs, which substantially increases both decoding complexity and memory requirements. Additionally, it can be observed that the PR code decoded with the double-bases decoder outperforms OSD(2) at FER levels below 10^{-4}.

V. Conclusion

This work proposed a double-bases BP decoding scheme for primitive rateless (PR) codes, which significantly improves the decoding performance using only two parity-check matrices (PCMs). By designing PR codes using primitive polynomials whose supports form a Golomb ruler, the first PCM is 4-cycle-free, while the second is generated adaptively based on channel log-likelihood ratio (LLRs) and kept fixed during decoding iterations. Simulation results show that the proposed method outperforms MBBP with lower complexity and offers competitive performance compared to polar and RM codes in the short block-length regime.

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